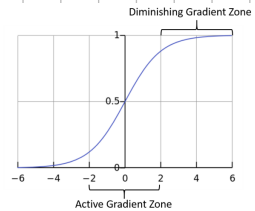


Logistic function $\rightarrow \sigma$
 call of well: logistic function

$$A = \frac{1}{1+e^{-x}}$$



$\sigma \rightarrow$ has this cool Property

①

$$\frac{d\sigma}{dt} = \sigma(1-\sigma)$$

logistic Model

It models as well the growing of a Population

①.1

σ Represents the size of the Population

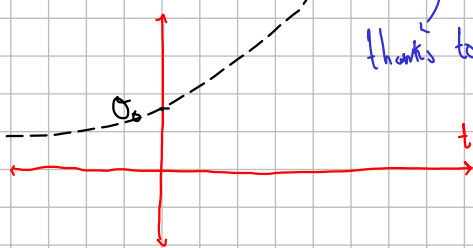
$$\frac{d\sigma}{dt} = r \cdot \sigma ; \quad \sigma(0) = \sigma_0$$

Int condition

Growing Rate

Solution = $\sigma(t) = \sigma_0 e^{rt}$

thanks to 'r' this grows infinitely ∞



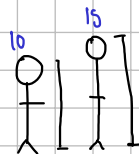
②

Better Model, not to ∞ $r = H(\sigma)$

$$\frac{d\sigma}{dt} = H(\sigma) \cdot \sigma$$

If we want to model the height of Humanity we want this 3 things

- $H(\sigma) \approx r$
- $H(\sigma) \approx 0 \rightarrow t \gg 0$
- $H(\sigma) < 0 \rightarrow t \gg 0$



$$H(\sigma) = (r - a\sigma)$$

$$\frac{d\sigma}{dt} = H(\sigma) \cdot \sigma$$

The rule of change of a Population is modeled:

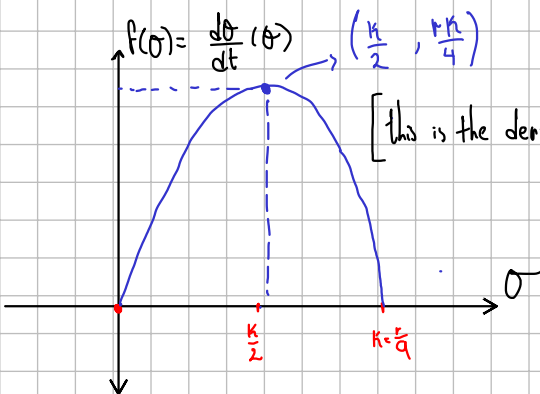
$$\frac{d\sigma}{dt} = (r - a\sigma) \cdot \sigma$$

Function that depends on σ

$$\frac{d\sigma}{dt} = f(\sigma)$$

$$\frac{d\sigma}{dt} = r(1 - \frac{\sigma}{K}) \cdot \sigma$$

$$\frac{d\sigma}{dt} = r(1 - \frac{\sigma}{K}) \cdot \sigma \quad \left[K = r/a \right]$$



[this is the derivative useful to analyze the real family functions]

with this derivative function we can know how the different functions are grow

$$\sigma(t) = ???$$

New class

Summary

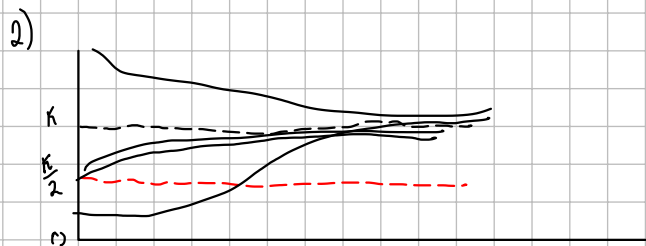
$$\frac{d\sigma}{dt} = (r - a\sigma)\sigma$$

$$\frac{d\sigma}{dt} = r(1 - \frac{\sigma}{K})\sigma; \quad K = \frac{r}{a}$$

$$\sigma(0) = \sigma_0 > 0$$

Condition

1) $r(1 - \frac{\sigma}{K})\sigma = f(\sigma)$



• Now we're gonna solve that equation:

$$\frac{d\sigma}{dt} = (r - a\sigma)\sigma$$

Rewrite

$$\frac{1}{(1 - \frac{\sigma}{K})\sigma} \cdot \frac{d\sigma}{dt} = r$$

$$\int \frac{1}{(1 - \frac{\sigma}{K})\sigma} d\sigma = \int r d\sigma + C$$

$$\int \frac{1}{(1 - \frac{\sigma}{K})\sigma} d\sigma = rt + C$$

$$\int \frac{A d\sigma}{1 - \frac{\sigma}{K}} + \int \frac{B}{\sigma} d\sigma = rt + C$$

Partial fractioning

[Partial fractioning]

$$A = \frac{1}{K}, \quad B = 1$$

$$\int \frac{\frac{1}{K} d\sigma}{1 - \frac{\sigma}{K}} + \int \frac{1}{\sigma} d\sigma = rt + C$$

$C \neq K$

$$-\ln|1 - \frac{\sigma}{K}| + \ln|\sigma| = rt + K$$

$$\ln \left| \frac{\sigma}{1 - \frac{\sigma}{K}} \right| = rt + K \rightarrow \frac{\sigma}{1 - \frac{\sigma}{K}} = K e^{rt}$$

$$\frac{\sigma}{1 - \frac{\sigma}{K}} = K e^{rt}$$

$$\sigma = K e^{rt} (1 - \frac{\sigma}{K})$$

$$\left(\frac{e}{K} e^{rt} + 1 \right) \sigma = C e^{rt}$$

$$\sigma(t) = \frac{C e^{rt}}{1 + \frac{C}{K} e^{rt}}$$

$$\sigma(t) = \frac{e^{rt}}{C e^{rt} \left(\frac{e^{rt}}{C} + \frac{1}{K} \right)}$$

$$\sigma(t) = \frac{1}{\frac{e^{-rt}}{C} + \frac{1}{K}}$$

$$\sigma(t) = \frac{C \cdot K}{C + K e^{-rt}} \xrightarrow{C=K} \sigma(t) = \frac{K \cdot K}{K + K e^{-rt}}$$

Logistic equation

• Now we're gonna find the Initial condition

$$\sigma_0 = \frac{C \cdot K}{C + K} \xrightarrow{\text{Process}} C = \frac{\sigma_0 K}{K - \sigma_0}$$

Here K is a parameter of the equation, K will be a number once we use a Real equation

this was our initial Problem

$$\frac{d\sigma}{dt} = r(1 - \frac{\sigma}{K})\sigma; \quad K = \frac{r}{a}$$

$$\sigma(0) = \sigma_0 > 0$$

Condition

to find the final solution we replace C in our logistic equation

$$\sigma(t) = \frac{C \cdot K}{C + K e^{-rt}} \xrightarrow{C = \frac{\sigma_0 K}{K - \sigma_0}}$$

$$\sigma(t) = \frac{\left(\frac{\sigma_0 K}{K - \sigma_0} \right) \cdot K}{\left(\frac{\sigma_0 K}{K - \sigma_0} \right) + K e^{-rt}}$$

For $\sigma_0 \neq K$

Final solution

$$\sigma(t) = \frac{\sigma_0 K}{\sigma_0 + (K - \sigma_0) \cdot e^{-rt}}$$

$$\sigma(t) = \frac{K}{1 + \left[\frac{K - \sigma_0}{\sigma_0} \right] e^{-rt}}$$

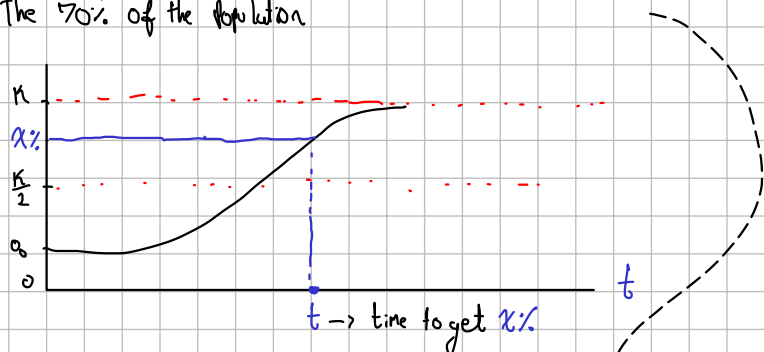
Final Result

$$\sigma(t) = \frac{\kappa}{1 + \left[\frac{\kappa - \sigma_0}{\sigma_0} \right] e^{-rt}}$$

r = growing ratio

κ = Max capacity of the Population

Now suppose that we need to know in what **time** we overcome the 50% of the population



How do we get there?

$$\sigma(t) = \frac{\kappa}{1 + \left[\frac{\kappa - \sigma_0}{\sigma_0} \right] e^{-rt}}$$

$$CT; \sigma(T) = \alpha \kappa?$$

The solution is

$$T = \frac{-\ln \left(\frac{1-\alpha}{\alpha} \cdot \frac{\sigma_0}{\kappa - \sigma_0} \right)}{r(\kappa - \sigma_0)}$$